

Yuzhuo Ao

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 Yuzhuo Ao |  Google Scholar

Hong Kong SAR, China

Research Interest

My research interests primarily focus on machine learning and computer vision. Specifically, I am particularly interested on systems for real-world modeling, representation and interaction.

Education

The Hong Kong University of Science and Technology

BSc in Data Science and Technology with Extended Major in AI (DSCT+AI)

2023/9 - 2027/9

Hong Kong SAR, China

École Polytechnique Fédérale de Lausanne

Exchange Program in Computer Science section

2026/2 - 2026/9

Lausanne, Switzerland

Nanyang Technology University

Summer Exchange Program

2025/06 - 2025/07

Singapore

Publications

- **SK-Adapter: Skeleton-Based Structural Control for Native 3D Generation** ECCV 2026
Anbang Wang*, Yuzhuo Ao*, Shangzhe Wu, Chi-Keung Tang
* Equal contribution.
- **ReasonNavi: Human-Inspired Global Map Reasoning for Zero-Shot Embodied Navigation** arXiv
Yuzhuo Ao*, Anbang Wang*, Yu-Wing Tai, Chi-Keung Tang
* Equal contribution.
- **XToM: Exploring the Multilingual Theory of Mind for Large Language Models** ACL 2026
Chunkit Chan, Yauwai Yim, Hongchuan Zeng, Zhiying Zou, Xinyuan Cheng, Zhifan Sun, Zheyang Deng, Kawai Chung, Yuzhuo Ao, Yixiang Fan, Cheng Jiayang, Ercong Nie, Ginny Y. Wong, Helmut Schmid, Hinrich Schütze, Simon See, Yangqiu Song

Projects

SK-Adapter: Skeleton-Based Structural Control for Native 3D Generation

2025/11 - 2026/03

Advised by Prof. Chi-Keung Tang and Prof. Shangzhe Wu

- Propose SK-adapter framework, a simple and yet highly efficient and effective framework that **firstly** unlocks precise skeletal manipulation for native 3D generation. SK-Adapter also allows for precise region-specific local 3D editing based on skeleton prompts.
- Construct the Objaverse-TMS dataset, comprising 24k high-quality text-mesh-skeleton triplets, addressing the data scarcity bottleneck in structure guided 3D generation

ReasonNavi: Human-Inspired Global Map Reasoning for Zero-Shot Embodied Navigation

2025/06 - 2025/09

Advised by Prof. Chi-Keung Tang And Prof. Yu-Wing Tai

- Developed ReasonNavi, a framework that leverages MLLMs for global reasoning and a deterministic planner for local navigation, creating a novel reason-then-act paradigm.
- Enabled unified, zero-shot navigation across diverse tasks (object, image, text goals), eliminating the need for task-specific fine-tuning or reinforcement learning.

Honors and Awards

HKUST Study Abroad Support (2025 Fall)
University’s Scholarship Scheme for Continuing Undergraduate Students (2025 Fall, 2024 Fall)
S. S. Chern Class Scholarship (for top math students in HKUST) (2024 Spring)
Dean’s List (2025 Fall, 2025 Spring, 2024 Spring, 2023 Fall)
University Admission Scholarship (2023 Summer)

Extracurricular Activity

32nd Executive Committee of Film Society, HKUSTSU	2025/05 - 2026/05
<i>Executive Committee Member</i>	<i>Hong Kong SAR, China</i>

Skills

Programming Languages	Python (Proficient), C/C++ (Proficient), R (Familiar)
Languages	Mandarin (Native), English (Proficient - IELTS 7.0 [09/2024])